Market Pulse

Step-by-step guide: push the code to GitHub, run it on your Mac, connect Zerodha for live BankNifty data, and use it every day.

What you are setting up

Market Pulse is a personal BankNifty analysis dashboard. It collects market data, the option chain and news, computes the expected range, key levels and buy/sell zones, and shows them at **http://127.0.0.1:8000**. It never places orders. Everything runs in Docker on your own machine.

Part	What you do	Where	Time
A	Push the code to a private GitHub repository	Windows PC	5 min
B	Install and start everything with one script	Mac	15-25 min
C	Connect Zerodha Kite for live prices (one-time setup)	Browser + Mac	10 min
D	Morning routine: log in to Zerodha, start live data	Mac	2 min / day
E	Optional: Claude (news + Ask) and Telegram alerts	Mac	10 min
F	Security rules, troubleshooting, command reference	-	reference

Practice data vs live data
Until you connect Zerodha, the dashboard plays back **practice data**: made-up prices tagged *synthetic*. It is for checking that everything works, not for trading. The Accuracy page refuses to show statistics for it, and alerts are not sent for it.

Before you start
Your **.env** file holds your passwords and API keys. It is never put on GitHub and never inside the Docker images. Copy it to the Mac yourself (USB stick or AirDrop) - never by email or chat.

Part A - Push the code to GitHub (Windows PC)

All the code is already committed in **D:\Aryan\Trading** on the **master** branch. These steps upload it to a private repository so the Mac can download it.

1 Create an empty private repository on GitHub

Open **https://github.com/new** and sign in.
Repository name: **market-pulse**. Visibility: **Private**.
Leave *Add a README*, *.gitignore* and *license* **unticked** - the repo must be empty.
Click **Create repository**.

2 Connect the local project to it

Open PowerShell and run (replace **<your-username>**):

```
cd D:\Aryan\Trading
git remote add origin https://github.com/<your-username>/market-pulse.git
git push -u origin master
```

3 Sign in when Git asks

A GitHub sign-in window opens the first time. Choose **Sign in with your browser** and approve. If you are asked for a password in the terminal instead, GitHub needs a **personal access token**, not your account password: GitHub > Settings > Developer settings > Personal access tokens > Fine-grained, give it access to **market-pulse** with *Contents: Read and write*.

4 Check what was uploaded

Refresh the repository page. You should see folders such as **packages**, **services**, **web**, **infra**, **scripts** and a README.
Make sure there is **no file called .env** in the list (only **.env.example**). If .env ever appears, delete the repository and tell me before doing anything else.

> **Later updates**
> After new changes on the PC, upload them with **git push**; on the Mac, download them with **git pull** and re-run the setup script.

Part B - Set up the Mac (one script)

The script **scripts/setup_mac.sh** installs everything that is missing and starts the app. It works on Apple Silicon and Intel Macs, and is safe to run again at any time.

1 Open Terminal

Press **Cmd + Space**, type **Terminal**, press Enter.

2 Install Homebrew and the GitHub tool (if you do not have them)

Skip the first command (two lines) if **brew --version** already works.

```
/bin/bash -c "$(curl -fsSL \
  https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
brew install gh
gh auth login
```

For **gh auth login** choose: GitHub.com, HTTPS, *Login with a web browser*, then enter the one-time code in the browser.

3 Download the project

```
gh repo clone <your-username>/market-pulse ~/market-pulse
cd ~/market-pulse
```

4 (Optional) Copy your .env from the PC

Copy **D:\Aryan\Trading\.env** into **~/market-pulse/** on the Mac (USB stick or AirDrop). If you skip this, the script creates a new one and asks for your keys.

5 Run the setup script

```
bash scripts/setup_mac.sh
```

It may ask for your **Mac password** (for Homebrew / Docker). Typing is hidden - that is normal.

What the script does, in order

#	Step	Notes
1	Installs missing tools: Homebrew, git, GitHub CLI, uv, Python 3.12, Docker Desktop	Already-installed tools are skipped
2	Starts Docker Desktop and waits for it	First time: **accept the licence** in the Docker window
3	Creates **.env** (readable only by you) with a random database password	Existing values are never overwritten
4	Asks for optional keys: Claude, Zerodha api_key / secret, Telegram	Typing hidden; press Enter to skip any
5	Installs the Python packages	
6	Builds practice data ending **today**	Playback starts 5 trading days back

#	Step	Notes
7	Builds and starts all 12 services, waits until healthy	First build: several minutes
8	Opens the dashboard in your browser	http://127.0.0.1:8000

If the script stops at Docker
On the very first run Docker Desktop shows a licence / sign-in window. Accept it (signing in is optional), wait until the whale icon in the menu bar stops animating, then run **bash scripts/setup_mac.sh** again. It continues where it left off.

You are done with Part B when...
The browser shows the BankNifty dashboard with an amber **SYNTHETIC DATA** banner (practice data), a green dot marked **open** at the top right, and panels marked **fresh** - data is flowing.

Part C - Connect Zerodha Kite (one-time setup)

Zerodha's **Kite Connect** API gives real-time prices and the option chain. It needs a Kite Connect developer app; market data requires the paid Kite Connect plan.

1 Open the Kite developer console

Go to **https://developers.kite.trade** and log in (it has its own account; signing up uses your email). Then click **Create new app** (type: Connect).

2 Fill in the app details

Field	Value
App name	Market Pulse (any name)
Zerodha Client ID	Your Zerodha login ID
Redirect URL	**http://127.0.0.1:8000/** (http, not https)
Postback URL	leave empty

If you already created the app (api_key starting **emz0...**), just open it and make sure the redirect URL is exactly **http://127.0.0.1:8000/**. With https you get *ERR_SSL_PROTOCOL_ERROR*.

3 Put the key and secret in .env on the Mac

From the app page copy the **API key** and **API secret**. If the setup script already asked you for them, skip this. Otherwise open the file:

```
cd ~/market-pulse
open -e .env
```

and fill in these two lines, then save:

```
KITE_API_KEY=your_api_key
KITE_API_SECRET=your_api_secret
```

Keep the API secret private
Never paste the secret into chat, email, screenshots or GitHub. If it was ever shared, click **Regenerate secret** on the app page and update .env.

Network requirement
Live data needs a network that can reach **api.kite.trade** directly. Your office network re-signs secure connections, which blocks it. Use home Wi-Fi or a phone hotspot. (Alternative: ask IT for the company root certificate and set **MP_CA_BUNDLE=/path/to/root.pem** in .env. Certificate checks are never switched off.)

Part D - Every morning: start live data

Zerodha access tokens expire every day at about **6 AM IST**, so you log in once each trading morning, ideally before 9:15.

1 Start the login

```
cd ~/market-pulse
uv run mp kite-login
```

It prints a link starting with **https://kite.zerodha.com/connect/login**.

2 Log in to Zerodha in the browser

Open the link, enter your Zerodha ID, password and 2FA. The browser then goes to **http://127.0.0.1:8000/?...request_token=...** - it may show the dashboard or an error page. Both are fine.

3 Paste the address back into Terminal

Copy the **whole address bar** and paste it at the **>** prompt, then press Enter. Do this within a few minutes - the request token works only once and expires quickly.
You should see *Logged in as ...* and *Saved KITE_ACCESS_TOKEN to .env*.

4 Check the connection

```
uv run mp kite-check
```

It prints your name, the live BankNifty price and the next expiries. If it fails, see Part F.

5 Switch the app to live data (first time only)

In .env set **MP_PROVIDER=kite** (it stays set). Then restart the services so they pick up today's token:

```
docker compose -f infra/docker-compose.yml --env-file .env up -d
```

6 Open the dashboard

Go to **http://127.0.0.1:8000**. The amber **SYNTHETIC DATA** banner is gone, the clock shows today's time, and prices update during market hours (9:15-15:30 IST).

Going back to practice data
Set **MP_PROVIDER=replay** in .env and run the same **docker compose ... up -d** command. To refresh the practice data so it ends today: **bash scripts/setup_mac.sh** again.

If MP_PROVIDER=kite but you have not logged in today
The market-data service stops with *set KITE_API_KEY and KITE_ACCESS_TOKEN* or *Kite rejected the access token* and the panels show **silent** or **stale**. Run steps 1-5 again.

Part E - Optional extras

Claude: news scoring and Ask Claude

With an Anthropic API key, news headlines are classified (sentiment, impact) and the **Ask Claude** panel explains the current setup. Claude only writes explanations - every number comes from the app's own calculations, and Claude cannot take any action.

1 Create a key

Go to **https://console.anthropic.com** > API keys > Create key.

2 Add it and restart

```
ANTHROPIC_API_KEY=sk-ant-...     # in .env
```

```
docker compose -f infra/docker-compose.yml --env-file .env up -d
```

Telegram alerts

Alerts go to your phone when price reaches a buy/sell zone, stop-loss or target. They are never sent for practice data.

1 Create a bot

In Telegram, message **@BotFather**, send **/newbot**, follow the steps and copy the **bot token**.

2 Find your chat ID

Send any message to your new bot. Then open **https://api.telegram.org/bot<token>/getUpdates** in the browser and copy the number after **"chat":{"id":**.

3 Add both and restart

```
TELEGRAM_BOT_TOKEN=123456:ABC...
TELEGRAM_CHAT_ID=123456789
```

```
docker compose -f infra/docker-compose.yml --env-file .env up -d
```

Part F - Security rules

Rule	Why
Never commit, email or screenshot **.env**	It holds every key and password. Git already ignores it.
Keep the GitHub repository **Private**	The code is personal; a public repo invites misuse.
Regenerate any secret you shared by mistake	Kite: Regenerate secret. Claude: delete the key. Telegram: /revoke in BotFather.
Open the dashboard only on **127.0.0.1**	All ports listen only on your own machine. Other hostnames and requests from other websites are rejected.
Do not disable certificate checks	On a blocked network, use another network or IT's root certificate.
Logs hide secrets automatically	Tokens, keys and passwords appear as <redacted> in Docker logs.

Troubleshooting

What you see	Cause	Fix
Panels say **silent** or **stale**	Market-data service stopped (no Kite token today, or the replay ended)	Part D, or set MP_PROVIDER=replay and restart
Dates on the dashboard are old	Practice data was built on an earlier day	Re-run **bash scripts/setup_mac.sh**
ERR_SSL_PROTOCOL_ERROR after Zerodha login	Redirect URL is https	Set it to **http://127.0.0.1:8000/** in the Kite console
kite-login: token is invalid or has expired	request_token used twice or too late	Run **uv run mp kite-login** again and paste quickly
kite-check: certificate / TLS error	Network intercepts secure connections	Use home Wi-Fi / hotspot, or MP_CA_BUNDLE
Kite rejected the access token	Token expired (after ~6 AM)	Part D steps 1-5
Docker did not start	Docker Desktop not finished first-run setup	Open Docker, accept, re-run the script
Port 8000 is already in use	Another app or an old copy is running	docker compose ... down, then up -d
Dashboard blank / not loading	Services still starting	Wait 30 s; check the status command below

Command reference (run inside ~/market-pulse)

Task	Command
Start / restart everything	`docker compose -f infra/docker-compose.yml --env-file .env up -d`
Stop everything	`docker compose -f infra/docker-compose.yml --env-file .env down`
Service status	`docker compose -f infra/docker-compose.yml --env-file .env ps`
Live logs of market data	`docker compose -f infra/docker-compose.yml --env-file .env logs -f ingest-market`
Overall health	`curl http://127.0.0.1:8000/api/health`
Zerodha daily login	`uv run mp kite-login`
Check Zerodha connection	`uv run mp kite-check`
Rebuild after git pull	`bash scripts/setup_mac.sh`
Fresh practice data only	`uv run mp synth --start YYYY-MM-DD --days 25 && uv run mp synth-news`

Addresses

What	Address
Dashboard	http://127.0.0.1:8000
API documentation	http://127.0.0.1:8000/api/docs
MCP servers (for Claude Desktop / Code)	http://127.0.0.1:8101/mcp (market), :8102 (options), :8103 (news), :8104 (signals)

This is an analysis tool, not financial advice and not an auto-trader. It never places orders. Always check levels yourself before trading.